From Machine Learning to Modern AI

Follow AI_USE_POLICY.pdf; the coding-assistant critique is taught in Session 21. This optional 20-minute orientation is an extension.

Optional 20-minute orientation within Session 23 in week eight, after learners finish their project rehearsal. It replaces an optional second rehearsal; it does not add a class, change marks or require a student account. If a learner still needs rehearsal support, prioritise that. This is an introduction to further study, not training in building a chatbot.

0–4 minutes — Connect to what you know

Artificial intelligence (AI) is the broad field of making computer systems perform tasks associated with intelligence. Machine learning is one approach: models learn patterns from examples. Data Analysis asks what the records show and how we can explain them; ML asks whether learned patterns can make useful predictions.

Point to a camp project. Ask: what were the inputs, the output and the check? A rent estimate is a numerical prediction (regression). A flower label is a category prediction (classification).

4–8 minutes — A neural network is another model family

A neural network combines numerical inputs through layers of learned calculations. It is not a miniature human brain. Deep learning uses networks with multiple layers. The learner already knows that fitting a model and checking its results are different steps; that principle still matters here.

For DA students, first define an input as information supplied to a calculation and an output as its result. No matrix arithmetic or model training is required in this orientation.

8–13 minutes — Predict a continuation

Write “The shop sold five …” and ask learners to suggest two endings. Both “books” and “pens” are plausible, but neither is evidence about a real shop. A large language model (LLM) is a model trained on large amounts of language data; it can generate text by predicting successive tokens. A token is a chunk of text, which may be a word or part of a word. This class activity is an analogy, not an implementation of an LLM.

Generative AI produces content such as text or images. Text generation does not by itself verify a claim. An optional tutor-only demonstration may use a prepared local model and synthetic text. No live service, download or learner login is needed: the activity works on paper.

13–18 minutes — Check a convincing claim

Display this explicitly teacher-written example, not a claimed model output:

“Book sales increased from 10 to 12 units. That is a 50% increase, proving the advertising worked.”

Ask students to check two things. The increase is 2 divided by 10, or 20%, not 50%. The two counts do not establish why sales changed. A fluent explanation can contain both a calculation error and an unsupported causal claim.

Show a repair: “Book sales increased from 10 to 12 units, a 20% increase. These counts alone do not tell us why.” Ask what further evidence would be needed before recommending more advertising.

18–20 minutes — Exit reflection and next steps

Each learner states one link between their project and modern AI, and one result they would independently check. Next study can include more Python, probability, linear algebra, neural networks and careful evaluation. Advanced models still require checks, sound data and clear limits.

Lecturer reference: Google’s introduction to machine learning connects regression, classification and generative systems. Use the distinctions above rather than describing every AI system as an LLM.